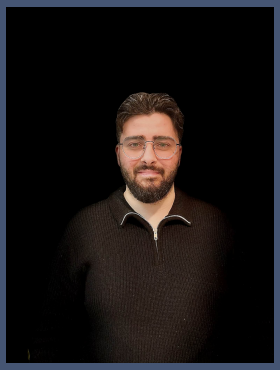
# Yousef **Mohammad**



## Profile

Final-year M.Sc. Software Engineering student at Linnaeus University and currently a Research Engineer / Full-Stack Developer. Experience building and deploying AI-enabled systems and full-stack web platforms with CI/CD. Interested in backend engineering, DevOps, security-aware development, and performance optimisation.

## Contact details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## Languages

English (Academic)
Swedish (Native)
Arabic (Native)

## Technical Skills

- **Languages:** Python, Java, C++, SQL
- **Web:** Vue.js, Nuxt, HTML/CSS, D3.js, Three.js
- **DB:** MySQL, PostgreSQL, MongoDB
- **DevOps:** Docker, CI/CD, GitHub, GitLab
- **AI/LLM:** RAG, LLM integration, semantic search
- **AI Coding Tools:** Cursor, VS Code Copilot, OpenCode, agent workflows
- **HPC:** CUDA (matrix multiplication optimisation)

## Experience

Research Engineer / Full-Stack Developer at *Linnaeus University (Växjö).* **2024.06–Present**

◇ Developing and deploying AI-driven and full-stack software systems across university projects.
◇ Prototyping LLM features (semantic search / ranking) and data visualisation components.
◇ Deploying via automated CI/CD pipelines; testing and iterating on architecture decisions.

Evening Shift Manager at *ICA Supermarket Mörners (Växjö).* **2013–Present**

◇ Operational leadership, staff coordination, and responsibility for service quality in evening shifts.

## Education

M.Sc. in Engineering: Software Engineering *Linnaeus University (Växjö).* **2021–2026 (Expected)**

◇ **Bachelor thesis:** Improving the Reliability of Educational AI Chatbots Using Retrieval-Augmented Generation.
◇ awaits.lnu.se — RAG-based student assistance system.

◇ **Master thesis (ongoing):** Reachability-Aware Vulnerability Prioritization in CI/CD using LLMs.
◇ Aggregating/prioritising findings from multiple tools using static reachability + repo context signals.

## Selected Projects

Inkuis — AI Event Aggregator Platform
◇ inkuis.lnu.se — LLM-based semantic search, filtering, and ranking for events.

AI Impact 2030 — Interactive Data Dashboard
◇ ai-impact-2030.onrender.com — D3.js/Three.js visualisations on generated data (HTML/CSS).

Multi-Target Compiler for Custom Language (OFP)
◇ Built a full compiler toolchain in Java using ANTLR: lexer/parser, symbol table, semantic analysis (type checking).
◇ Dual-target codegen: Python transpilation + JVM bytecode via ASM; supports functions, recursion, arrays, control flow, and multiple types.

Misty Robotics — LLM Conversational System
◇ Integrated LLM-driven multi-turn dialogue on Misty; orchestration layer for prompts, tool/agent calls, parsing, and dialogue state.

CUDA Matrix Multiplication Optimisation
◇ Implemented GPU matrix multiplication and improved performance via parallelisation and memory access optimisation.